Week 2 Deliverable: Stakeholder analysis, SRS And Use case model

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1. Introduction

This document contains the stakeholder analysis and conflict resolution, the Software Requirements Specification (SRS), and the Use Case Model. This phase specifies what the LMS must do and how its behavior will be verified.

The Phase 1 choices remain unchanged: a simple C++17 desktop GUI, local SQLite persistence, a single school computer deployment for v1.0, no Internet dependency for normal operation, and a second laptop for independent deployment testing only.

The stakeholder statements below were supplied in the case study. The conflict resolutions are the project decisions used to make the requirements consistent, they are not presented as the results of new interviews or as approval already obtained from the school or supervisor.

2. Stakeholder Analysis and Conflict Resolution

2.1 Stakeholder interests

Stakeholder	Interest	Requirement / concern
Principal	School oversight, straightforward assessment procedures	Clear interface for non-technical teachers, no student access to marks before HoD review
Administrator	Maintenance of school records and accounts	Manage users, departments and courses, assign lecturers, enroll students
Lecturer	Creation and marking of coursework	Create assignments and deadlines, grant individual extensions, record marks and written feedback
Head of Department	Quality control of departmental results	Review/moderate departmental marks and release only approved results, late-work penalty policy
Student	Access to coursework and personal results	View enrolled courses, submit/resubmit work and view released marks and written feedback
Student Representative	Student interests and assessment fairness	Allow correction through resubmission, provide written feedback, do not reject late work outright
IT Officer	Reliability and protection of school data	Internet-independent operation, password protection, responsive use with 800 student records and older hardware

2.2 Conflicts, ambiguities and selected resolutions

ID	Issue	Resolution adopted for the SRS
C1	Lecturer wants immediate marks, Principal and HoD require checking before visibility.	Marks remain unreleased after lecturer marking. HoD moderates and explicitly releases each approved result.
C2	HoD sets penalties, students request late work not be rejected.	Accept first-time late submissions, deduct 10% of maximum mark per commenced late day for up to three days, more than 72 hours late has final mark zero.
C3	Lecturer can extend the deadline, but penalty effect is unclear.	An approved individual extension changes the student-specific effective deadline for late calculations.
C4	Resubmission is allowed, but the marked version is unspecified.	Retain submission history and mark the latest valid version submitted before the effective deadline.
C5	It is unclear whether results must be released together for a course.	Moderation and release occur per assessment result, no unapproved result becomes visible.
C6	The instruction to keep the system simple is subjective.	Use clear role-specific GUI screens and a measurable usability acceptance test (NFR-04).
C7	"Quickly with all 800 students on it" does not specify concurrent sessions.	Use a database with 800 stored student accounts for the performance test, not an assumed 800 concurrent sessions.
C8	"Passwords must be protected" does not specify a mechanism.	Use a recognized library to store salted password hashes, never store plain-text passwords.
C9	"Work offline" may be mistaken for a prohibition on all networking.	Operate without Internet access, v1.0 is a single-computer installation, as selected in Phase 1.
C10	The level of detail for mark-change history is unclear.	Log responsible user, timestamp, old/new marks and reason when an existing mark changes, preserve the audit history.
C11	The brief does not define what form student work takes.	Support typed text, file attachments, or both in the same submission, require at least one form of content.

Rules C2-C5 and C11 are specific project interpretations of gaps in the brief. Their detailed boundaries are documented in the SRS.

3. Software Requirements Specification (SRS)

3.1 Purpose, users and scope

The LMS manages user accounts, departments, courses, assignment publication, submissions, late penalties, lecturer marking, HoD moderation and the release of results. Direct system users are Student, Lecturer, Head of Department and Administrator. The principal, student representative and IT officer inform requirements but are not additional LMS actor roles.

3.2 Operating environment and constraints

The application shall use C++17 with a simple desktop GUI and SQLite storage on a single school computer for v1.0. It shall not require Internet connectivity for normal use. SQLite access will be isolated behind a data access layer during implementation. A second laptop will be used for deployment verification but is not part of the operational architecture.

3.3 Identification and MoSCoW priorities

Every functional requirement has a stable identifier FR-01 to FR-18. Every non-functional requirement has identifier NFR-01 to NFR-06. Must = essential to the required first release, Should = expected but lower priority than a Must. Could and Won't are valid MoSCoW categories but no optional features have been added merely to populate those categories. The table of verification methods defines how each requirement can be checked, not a claim that tests have been executed.

3.4 Functional requirements

ID	Requirement (the system shall...)	Priority	Verification method
FR-01	Allow registered users to log in with username and password and show the dashboard for their role.	Must	Valid/invalid login tests for all four roles.
FR-02	Enforce role-based permissions so users can perform only actions assigned to their role.	Must	Attempt authorized and unauthorized actions in each role.
FR-03	Allow the Administrator to create, view, update and deactivate user accounts.	Must	Create, retrieve, modify and deactivate test accounts.
FR-04	Allow the Administrator to create, view, update and manage departments.	Must	Create and update department records, verify persistence.
FR-05	Allow the Administrator to create, view, update and manage courses associated with departments.	Must	Create/update courses and verify department links.
FR-06	Allow the Administrator to assign lecturers to courses.	Must	Assign a lecturer and check access to that course.
FR-07	Allow the Administrator to enroll students in courses.	Must	Enroll a student and verify their enrolled-course list.
FR-08	Allow a Lecturer to create assignments for assigned courses with a title, deadline and maximum mark.	Must	Create an assignment and inspect saved details.
FR-09	Allow a Lecturer to grant an individual assignment deadline extension to an enrolled student.	Must	Grant extension and verify the student-specific effective deadline.
FR-10	Allow students to view their enrolled courses and associated assignments.	Must	Check assigned and unassigned course visibility.
FR-11	Allow an enrolled Student to submit typed text, a local file attachment, or both, record student, assignment, content and timestamp.	Must	Submit text only, file only and both, verify content and timestamp.
FR-12	Allow a student to resubmit revised text and/or attachments before the effective deadline while preserving earlier versions.	Must	Resubmit before and after effective deadline, inspect version history.
FR-13	Accept late first submissions and deduct 10% of the maximum mark per commenced 24-hour period late, up to 72 hours, more than 72 hours late gives a final mark of zero.	Must	Boundary tests at deadline, 24h, 48h, 72h and just beyond 72h.
FR-14	Allow a Lecturer to view submissions for assigned courses, enter a valid mark and provide written feedback.	Must	Record feedback/mark, check range and role restrictions.

ID	Requirement (the system shall...)	Priority	Verification method
FR-15	Allow an HoD to review and moderate marks belonging to the HoD's department.	Must	Moderate departmental mark, deny access to other departments.
FR-16	Allow an HoD to release a mark only after the required moderation approval.	Must	Reject unapproved release, accept approved release.
FR-17	Allow a Student to view the student's own released marks and feedback but not unreleased results.	Must	Check visibility before and after release and across accounts.
FR-18	For every modification of an existing mark, record user, timestamp, previous value, new value and reason in an audit log.	Must	Change marks and inspect complete audit entries, test failed writes.

3.5 Non-functional requirements

ID	Property	Measurable requirement	Priority	Verification
NFR-01	Performance	At least 19 of 20 measured valid login attempts shall complete within 2 seconds when the database holds 800 stored student accounts on the target computer.	Must	Record 20 login durations and count those at or below 2 seconds.
NFR-02	Offline availability	All required core user workflows shall be completed while the school computer has no Internet connection.	Must	Disconnect Internet and execute the predefined core acceptance workflows.
NFR-03	Password security	100% of stored passwords shall be represented by salted cryptographic hashes produced by a recognised password hashing library, no plain-text passwords shall be stored.	Must	Inspect password records and the credential-storage implementation.
NFR-04	Usability	Four role testers, one per role, shall each complete a predefined common task without help after no more than 15 minutes of introduction.	Must	Record training time, completion and assistance for four role-specific tasks.
NFR-05	Accessibility	All primary GUI operations for each of the four roles should be executable through keyboard navigation without a mouse.	Should	Execute every primary role-specific workflow using keyboard input only.
NFR-06	Persistence reliability	All 20 successfully committed test records covering submissions, marks and audit entries shall remain retrievable and unchanged after closing and restarting the application.	Must	Save 20 known records, restart, and verify content and values match.

The numerical values in NFR-01 and NFR-04 to NFR-06 are proposed acceptance targets. The case study supports the need for responsiveness, simplicity and reliability but does not report that these targets have already been achieved. No benchmark or usability trial is claimed at this stage.

3.6 Business rules

Rule	Defined behavior
BR-01	A lecturer-saved result remains unreleased until the authorised HoD reviews approves it for release.
BR-02	Late periods are counted as commenced 24-hour periods after the effective deadline: (0,24) hours = 1 day, (24,48) = 2, (48,72) = 3, more than 72 hours = zero final mark. For 1-3 periods, deduct 10% of assignment maximum from earned mark per period, final mark cannot fall below zero.
BR-03	An approved individual extension replaces the original assignment deadline for that student when classifying lateness and resubmissions.
BR-04	The latest valid version submitted on or before the effective deadline is used for marking, earlier versions remain retained. A first-time late submission can still be recorded, but a late replacement of an existing submission is not permitted.
BR-05	Every modification to an existing mark records the user, time, old value, new value and reason, unaudited changes must not be committed.
BR-06	User access is limited by role and course/department enrolment or assignment, and students may see only their own released results.
BR-07	A submission must contain non-empty typed text, a readable and storable file, or both. A failure to store either the submission record or its required content cannot be reported as successful.
BR-08	The HoD may approve or adjust a mark only within the applicable allowed range. Any adjustment is audited, release requires prior moderation approval.

3.7 Assumptions and items still require confirmation

ID	Open detail	Required follow up
A1	Commenced 24-hour periods are used for late penalties, the brief does not explicitly say how partial days count.	Gate 1 confirmation
A2	The latest valid resubmission replaces earlier versions for marking, but earlier versions remain stored.	Gate 1 confirmation
A3	Typed text and attached files are supported, allowed file types, size limits and precise storage representation are not specified in the brief.	Set before design is approved
A4	A mark may be moderated and released individually rather than waiting for all marks in a course.	Gate 1 confirmation
A5	800 refers to student accounts stored for the acceptance test, no requirement for 800 concurrent active sessions is stated.	Gate 1 confirmation
A6	Detailed hardware capabilities, tester availability and relevant institutional data-retention policy are not supplied.	Verify when test/deployment plans are prepared

4. UML Use Case Model

4.1 UML Use Case Diagram

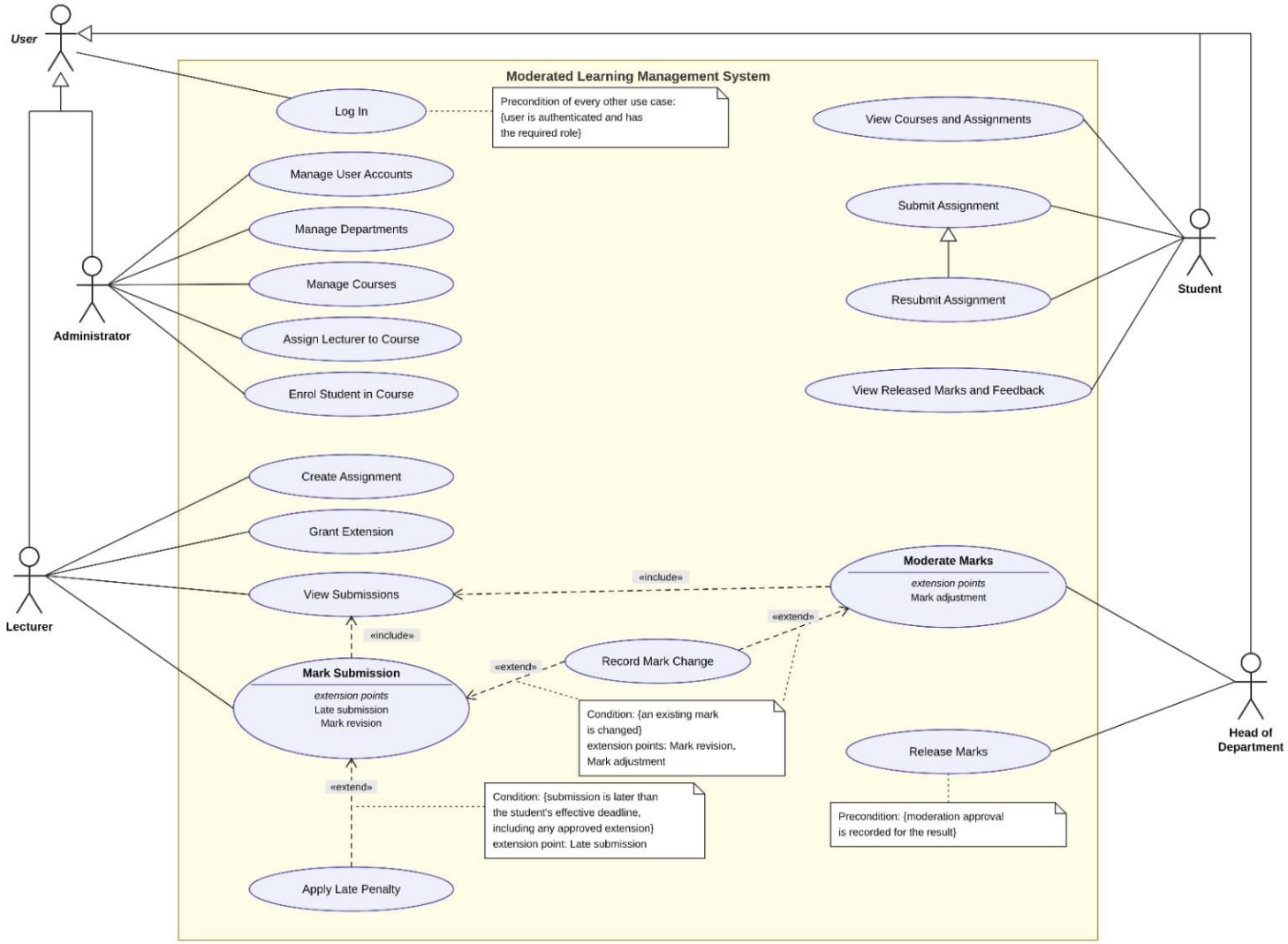


Figure 1. LMS use case diagram. The roles are outside the system boundary, use cases are inside it.

4.2 Actors and supported goals

Actor	Main use cases
Administrator	Manage User Accounts, Manage Departments, Manage Courses, Assign Lecturer to Course, Enroll Student in Course
Lecturer	Create Assignment, Grant Extension, View Submissions, Mark Submission
Head of Department	Moderate Marks, Release Marks
Student	View Courses and Assignments, Submit Assignment, Resubmit Assignment, View Released Marks and Feedback

Log In is a shared use case for all roles. The diagram focuses on user goals and does not treat SQLite, GUI controls, or data-access classes as external actors. The core exception paths are specified in the accompanying use case descriptions.

4.3 Relationship explanations

Relationship	Use in the model	Meaning
«include»	Mark Submission → View Submissions	The lecturer views the submitted work while marking.
«include»	Moderate Marks → View Submissions	The HoD reviews the submission during moderation.
«extend»	Apply Late Penalty → Mark Submission [late]	The late penalty is applied only when the submission is late.
«extend»	Record Mark Change → Mark Submission; Moderate Marks [existing mark changed]	The audit entry is mandatory whenever an existing mark is changed.
Generalisation	Resubmit Assignment → Submit Assignment	Resubmission is a specialised submission; previous versions are retained.
Actor generalisation	Administrator, Lecturer, Student, HoD → User	All four roles specialise User and share its Log In association.
Association	Each actor ↔ relevant use cases	The actor participates in the permitted functions.

5. Fully Dressed Use Case Specifications

5.1 UC-01: Submit Assignment

Field	Specification
Primary actor	Student
Related FR	FR-02, FR-07, FR-10, FR-11, FR-12, FR-13
Trigger	Student selects Submit Assignment for an assignment.
Preconditions	Student is logged in and enrolled in the course. A relevant assignment exists and is available for submission.
Success postcondition	Timestamped submission version with text and/or file is saved for the correct student/assignment, status records whether it is late.
Failure postcondition	No successful version is created, any previously saved version remains unchanged.
Related rules	BR-02, BR-03, BR-04, BR-06, BR-07

Main success flow

1. Student opens an enrolled course and selects an assignment.

2. System shows the assignment, original deadline and effective deadline (including any approved individual extension).

3. Student enters text, chooses a local file, or provides both.

4. Student selects Submit.

5. System validates account permissions, enrolment, assignment and the presence/readability of submission content.

6. System saves the version, text/file content and submission timestamp.

7. System classifies the submission as on-time or late against the effective deadline.

8. System confirms successful submission and displays the recorded timestamp.

Alternative flows

A1 - Text only: At step 3, the student supplies non-empty text without a file.

A2 - File only: At step 3, the student supplies a readable local file without typed text.

A3 - Both: At step 3, typed text and a file are stored as one submission version.

A4 - First submission is late: At step 7, system records lateness, penalty is calculated when marking takes place.

A5 - Individual extension: At step 2, the approved extended deadline is applied.

A6 - Resubmission: A prior version exists and the effective deadline has not passed, a new version is saved while older versions remain.

Exception flows

E1 - Empty work: At step 5, reject when neither text nor file is present.

E2 - File cannot be read or stored: At step 5 or 6, show error and record no successful version.

E3 - Unauthorised course: At step 5, deny a submission from an unenrolled student.

E4 - Late resubmission: A replacement attempted after the effective deadline is rejected, the existing version remains unchanged.

E5 - Storage error: At step 6, roll back partial submission data and do not display a success confirmation.

5.2 UC-02: Mark Submission

Field	Specification
Primary actor	Lecturer
Related FR	FR-02, FR-06, FR-13, FR-14, FR-18
Trigger	Lecturer selects a submitted assignment in an assigned course.
Preconditions	Lecturer is logged in and assigned to the course, the selected submission exists, assignment has a maximum mark.
Success postcondition	Lecturer result and written feedback are saved, result remains unreleased for HoD moderation, any revision is audited.
Failure postcondition	Invalid or unauthorised changes are not saved, previous values remain intact.
Related rules	BR-01, BR-02, BR-03, BR-05, BR-06, BR-08

Main success flow

1. Lecturer opens an assigned course and selects an assignment.
2. System lists student submissions and their timestamps and late statuses.
3. Lecturer opens submitted text and any attached file.
4. Lecturer enters an earned mark within the maximum assignment and written feedback.
5. System validates the mark and computes lateness relative to the student-specific effective deadline.
6. System applies any required late penalty or the more-than-three-days zero rule.
7. System saves the assessed result and feedback as unreleased.
8. If an existing mark is modified, system stores the prior value, new value, user, timestamp and supplied reason in the audit log.
9. System confirms the result is saved for moderation.

Alternative flows

A1 - On time or extended: No penalty is applied.

A2 - Up to 72 hours late: Deduct 10% of maximum available mark per commenced 24-hour period, clamp at zero.

A3 - More than 72 hours late: Store final mark as zero while retaining the submission and feedback.

A4 - Revision: Require a reason for updating an existing mark and record the change in the audit log.

Exception flows

E1 - Unassigned lecturer: Deny access to the course or submission.

E2 - Invalid mark: Reject values outside 0 to the assignment maximum.

E3 - Missing submission: Display error without recording a mark.

E4 - Storage/audit failure: Reject or roll back update rather than creating an unaudited change.

5.3 UC-03: Moderate and Release Marks

Field	Specification
Primary actor	Head of Department
Related FR	FR-02, FR-15, FR-16, FR-17, FR-18
Trigger	HoD selects an assessment requiring moderation.
Preconditions	HoD is logged in, the assessment belongs to the HoD's department, and a lecturer’s result exists.
Success postcondition	The moderated, approved result is released to the correct student, and any mark adjustment has a recorded audit entry.
Failure postcondition	Unapproved results remain unreleased, valid earlier data and audit history remain intact.
Related rules	BR-01, BR-05, BR-06, BR-08

Main success flow

1. HoD opens the moderation list for the HoD's department.
2. System shows recorded but unreleased assessment results in that department.
3. HoD selects an assessment and reviews the submission, mark, feedback and any late-rule outcome.
4. HoD approves the mark or adjusts it within permitted limits and supplies a reason if changing it.

5. System validates the decision and records any change to the mark in the audit log.

6. System marks the assessment as moderated and approved.

7. HoD selects Release.

8. System checks moderation approval and releases the mark and written feedback to the correct student.

9. System confirms release.

Alternative flows

A1 - Unchanged mark: HoD approves the lecturer mark without adjustment.

A2 - Adjusted mark: HoD changes mark within allowed bounds and records a reason, the change is audited.

A3 - Not approved: HoD withholds approval and the result remains hidden for correction/review.

Exception flows

E1 - Wrong department: Deny access and moderation.

E2 - Missing/invalid result: Prevent approval or release until corrected.

E3 - Premature release: Reject release when moderation approval has not been recorded.

E4 - Persistence/audit failure: Do not claim success, prevent partial changes and preserve unreleased state.

6. Initial Requirements-to-Use-Case Traceability

The table below links all 18 functional requirements to named use cases. Source classes, database tables, code files and test IDs will be added in later phases, those artefacts have not yet been designed or implemented.

Requirement	Use cases satisfying requirement
FR-01	Log In
FR-02	All authenticated role-specific use cases
FR-03	Manage User Accounts
FR-04	Manage Departments
FR-05	Manage Courses
FR-06	Assign Lecturer to Course
FR-07	Enroll Student in Course
FR-08	Create Assignment
FR-09	Grant Extension
FR-10	View Courses and Assignments
FR-11	UC-01 Submit Assignment
FR-12	Resubmit Assignment, UC-01
FR-13	UC-01 (late classification), UC-02 (penalty)
FR-14	View Submissions; UC-02 Mark Submission
FR-15	Moderate Marks (UC-03)
FR-16	Release Marks (UC-03)
FR-17	View Released Marks and Feedback; UC-03 (release)
FR-18	Record Mark Change (UC-02 and UC-03)

7. Gate 1 Consistency and Validation Check

Check	Expected	Result / note
Required scope	Four user roles, coursework, marking, HoD moderation and release	Covered by stakeholder, FR and UC sections.
Counts and IDs	Exactly 18 FR and 6 NFR, IDs unique	FR-01..FR-18, NFR-01..NFR-06.
Verification	Every requirement has test or inspection criterion	Defined in SRS requirement tables.
Submission media	Typed text, attached file, or both	FR-11/12, BR-07 and UC-01 agree.
Penalties and extensions	Late rule applies against effective deadline	FR-09/12/13, BR-02/03/04, UC-01/02 agree.
Mark confidentiality	No student view until HoD approves and releases	FR-15/16/17, BR-01 and UC-02/03 agree.
Audit trail	All changes to existing marks are traceable	FR-18, BR-05, UC-02/03 agree.
Phase 1 baseline	Desktop GUI, local SQLite, Internet-independent, one computer	No conflicting deployment requirement.
Open questions	Partial-day penalty, file types/sizes, testing arrangements	Listed as decisions needing confirmation, no claim of stakeholder approval.

-Abbreviations Key

LMS – Learning Management System

UC – Use Case

BR – Business Rule

MoSCoW – Must Have, Should Have, Could Have, Won't Have

CR1 – Change Request 1

FR-01 to FR-18 – Functional Requirement identifiers

NFR-01 to NFR-06 – Non-Functional Requirement identifiers

UC-01 to UC-03 – Fully dressed Use Case identifiers

BR-01 to BR-08 – Business Rule identifiers

C1 to C11 – Stakeholder conflict and ambiguity identifiers

A1 to A6 – Assumptions requiring confirmation

8. References

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